

# Kunal Maheshwari

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## Summary

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**Java Software Engineer** with 4.6+ years developing and maintaining secure, stable enterprise systems across **healthcare payer integrations** and **automotive configuration and recommendation products**.

**Delivery scope:** Carries application changes from requirements and system design through Java/Spring development, REST/SOAP and SQL integration, automated testing, CI/CD validation, release support, and operational stability in large corporate environments.

**Engineering approach:** Troubleshoots upstream and downstream behavior across application, database, messaging, and external-service boundaries; uses testing, observability, deterministic validation, and performance analysis to improve resiliency and maintainability.

**Working style:** Collaborates across Agile engineering, QA, database, deployment, vendor, and business-facing teams; communicates technical implications, validates changes through peer review and automated testing, documents solutions, and mentors developers.

## Technical Skills

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**Languages:** Java 8/11; Core Java (Collections, Streams, Lambdas, Concurrency); SQL; Python.

**Backend/Integration:** Spring Boot, Spring Framework, Spring MVC, REST/SOAP APIs, JAX-WS, Java EE/J2EE, EJB, JSF, Hibernate/HQL, JPA, JDBC/OJDBC, SOA, JMS, Azure Service Bus, Azure Relay, Azure Functions, IBM WebSphere, Oracle WebLogic.

**AWS & Cloud Projects:** AWS S3, Lambda, EFS, ECS, RDS, CloudWatch, IAM, VPC; Terraform, Docker, PostgreSQL/JDBC, event-driven ingestion, restart checkpoints; Microsoft Azure.

**DevOps, Testing & Operations:** Jenkins, Maven, IBM UrbanCode, GitLab CI/CD, GitHub Actions, Git, SVN; JUnit/JUnit4, Mockito, Integration Testing, Log4j2, SoapUI, Postman; application resiliency, production support, observability, root-cause analysis.

**Frontend & Applied AI Projects:** React, TypeScript, Vite, Tailwind CSS; Python, FastAPI, PyTorch, Hugging Face Transformers, T5, PEFT/LoRA, offline ROUGE evaluation.

**Engineering Practices:** OOP, Design Patterns, Agile/Scrum, System Design, Secure Change Validation, Performance Tuning, Technical Documentation, Code Reviews, Mentoring.

## Work Experience

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**NTT Data**, Bengaluru, India

*Software Developer, Aug 2022 - Aug 2024*

- Developed, tested, and maintained **Java 8/11, JAX-WS/Spring, and REST/SOAP** healthcare integration services on IBM WebSphere, supporting **50K+ daily requests at 99.99% SLA availability**.
- Integrated hybrid service flows across **Azure Relay, Azure Service Bus, Azure Functions, and JMS**, synchronous downstream REST calls, and Oracle/OJDBC processing, accounting for upstream/downstream implications with explicit timeout and partial-failure handling.
- Integrated Java services with **Oracle through OJDBC**, consuming established views and PL/SQL procedures while applying service-layer member/provider business rules.
- **Led a project-wide Log4j2 migration**, owning the initial PoC, preserving production logging contracts, documenting a repeatable upgrade path, and reviewing the team rollout.
- Integrated an AOP-based service-monitoring library that propagated correlation IDs and captured **latency, success/failure, and request-volume telemetry** across services.
- Supported Jenkins/Maven/IBM UrbanCode build and deployment workflows, JUnit/SoapUI validation, and post-release smoke/business checks; contributed go/no-go/rollback recommendations and production root-cause investigations.
- Supported root-cause analysis of a **WebSphere memory/resource-retention incident**, contributing code-path review, mitigation strategy, and rolling-restart validation while maintaining operational stability.

Capgemini, Mumbai, India

Senior Analyst, Jul 2020 - Aug 2022

- Developed and supported a Java EE/J2EE, JSF, Hibernate/HQL enterprise application on Oracle WebLogic with DB2 for Mercedes/Daimler Vans configuration, recommendation, and ordering workflows.
- Integrated a Python vehicle-configuration prediction workflow with Java services, then enforced deterministic option/package validation before recommendations were exposed to dealer users.
- **Reduced VIN repeat-configuration API latency from 60+ seconds to 10–20 seconds** by moving stable option-code mapping from Java into set-based HQL/DB2 CASE expressions while retaining business validation in Java.
- Diagnosed cross-system issues by analyzing Python output, Java validation logs, and DB2 reference data; used diverse operational data to isolate defects and supported Jenkins/GitLab CI/CD, JUnit4, code reviews, and production debugging.
- **Mentored junior developers** through codebase walkthroughs, incremental defects and features, debugging support, and code reviews.

Capgemini, Mumbai, India

Analyst Trainee, Jan 2020 - Jun 2020

- Built Java Spring/Spring MVC, Angular, and PostgreSQL modules for fitness-club and digital-wallet applications, implementing authentication, email/OTP login, CRUD, account balances, transactions, and profile workflows.

## Projects

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### AWS Automated Spark Pipeline

[github.com/preferablehuman/AWS-automated-spark-pipeline](https://github.com/preferablehuman/AWS-automated-spark-pipeline)

- Built an **event-driven AWS data pipeline** where S3 uploads trigger Lambda staging to EFS, an ECS-hosted Docker workload processes micro-batches, and results persist to RDS PostgreSQL over JDBC.
- Provisioned VPC, IAM, S3, Lambda, EFS, ECS, RDS, security groups, and CloudWatch resources with Terraform; processed an **8GB+ dataset** and persisted checkpoints for restart recovery.

### Privacy-Preserving AI Email Client (Graduate Capstone)

- Built an LLM-powered email summarization and classification system with a **FastAPI model service, PyTorch, Hugging Face Transformers, T5, and PEFT/LoRA**, plus Docker, MinIO, Prometheus, React, and TypeScript.
- Implemented diagnostics and an offline evaluation workflow; critically assessed model outputs using repeatable metrics of **ROUGE-1 F 53.04, ROUGE-2 F 44.90, and ROUGE-L F 45.90**.

## Education

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Central Michigan University, Mount Pleasant, MI, USA

Aug 2024 - Dec 2025

Master of Science in Computer Science, GPA: 3.87/4.0

Lovely Professional University, Punjab, India

Aug 2016 - May 2020

Bachelor of Technology in Computer Science and Engineering, CGPA: 8.02/10

## Certifications

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Microsoft Certified: Azure Fundamentals | Azure Developer Associate | Azure Data Engineer Associate